sign-tx Signing Benchmark Report

Cosmian KMS 5.27.1 · PKCS#11 · NIST P-256

2026-09-09

Summary

`sign-tx` was benchmarked against a real Cosmian KMS (v5.27.1) over PKCS#11, signing Solana-shaped transfer transactions with a NIST P-256 key created via `ckms`. Ed25519 could not be benchmarked against this KMS because of an upstream bug: Cosmian's PKCS#11 provider does not expose Ed25519 private keys created through `ckms`/KMIP (filed as Cosmian/kms#1183). P-256 support was added to `sign-tx` as a permanent feature (`--curve p256`) to work around this and obtain real, KMS-backed signing latency numbers.

Headline result: ~2.8 ms median (p50) end-to-end signing latency, ~355 signatures/sec sequential throughput. The KMS network round trip (`C_Sign`) accounts for ~90% of total time.

Test Configuration

Parameter	Value
KMS	Cosmian KMS 5.27.1 (non-FIPS), PostgreSQL backend
Transport	Cosmian PKCS#11 provider (`libcosmian_pkcs11.so`)
Curve / mechanism	NIST P-256, `CKM_ECDSA` (DER-encoded, over SHA-256 digest)
Key provisioning	`ckms ec keys create --curve nist-p256`

Parameter	Value
Payload	Solana transfer message, 150 bytes
Iterations	200 (+ 20 warmup)
Concurrency	1 (sequential — latency, not throughput under load)
Environment	Docker Compose (KMS + PostgreSQL + bench client)

Results

Phase	n	min	p50	p90	p99	p999	max	stddev
`C_SignInit`	200	2.60 µs	6.37 µs	9.54 µs	19.50 µs	29.82 µs	29.82 µs	3.47 µs
`C_Sign` (network)	200	1.204 ms	2.480 ms	3.236 ms	4.060 ms	5.632 ms	5.632 ms	641.27 µs
verify (local)	200	141.15 µs	279.66 µs	332.35 µs	379.22 µs	397.25 µs	397.25 µs	53.88 µs
total	200	1.348 ms	2.819 ms	3.541 ms	4.411 ms	5.929 ms	5.929 ms	660.82 µs

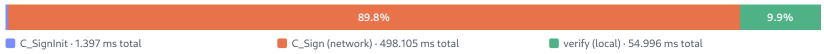
- **Throughput:** 354.7 sig/s at p50, 360.7 sig/s at mean
- **`C_Sign` share:** 89.8% of total time (the network-bound component)
- **Wall clock:** 554.828 ms for 200 iterations
- **Warmup (discarded):** first = 2.415 ms, p50 = 2.168 ms, last-min = 1.622 ms

Chart

Cosmian KMS PKCS#11 provider — 200 signatures via PKCS#11

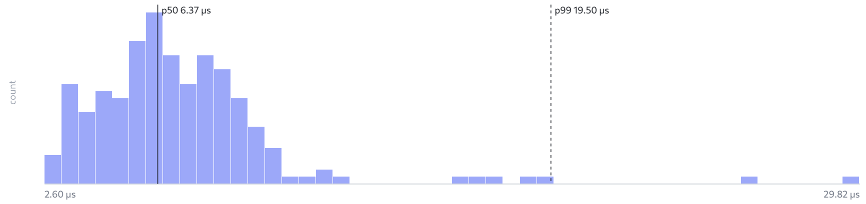
Per-step signing latency · sequential · every sample retained

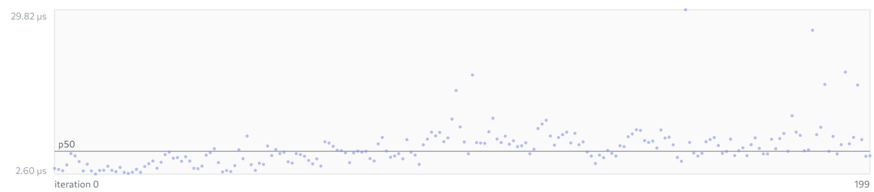
Where the time goes



C_SignInit

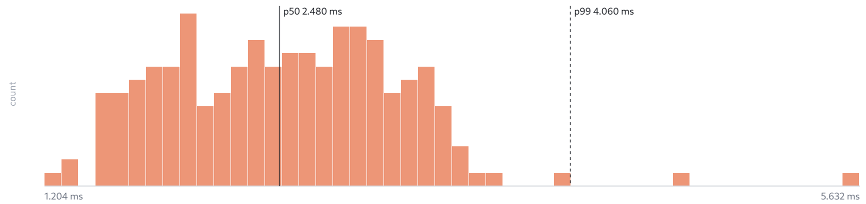
n=200 · p50 6.37 µs · p99 19.50 µs · max 29.82 µs

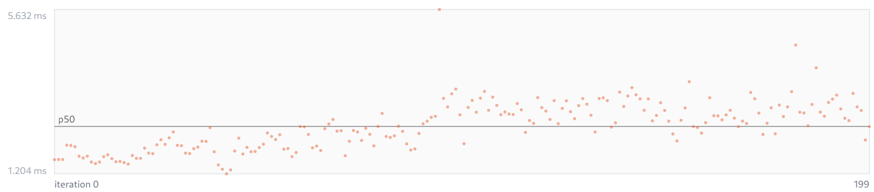




C_Sign (network)

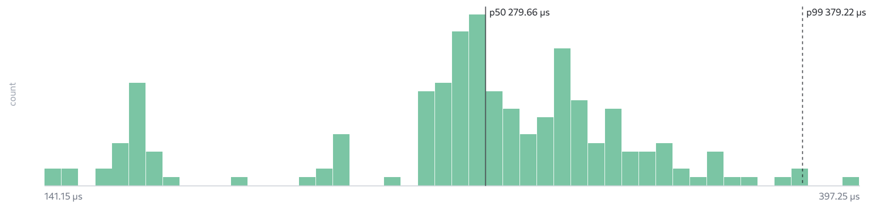
n=200 · p50 2.480 ms · p99 4.060 ms · max 5.632 ms

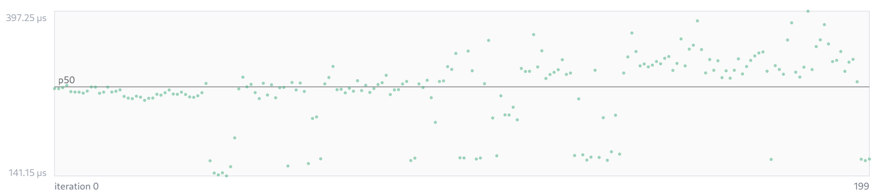




verify (local)

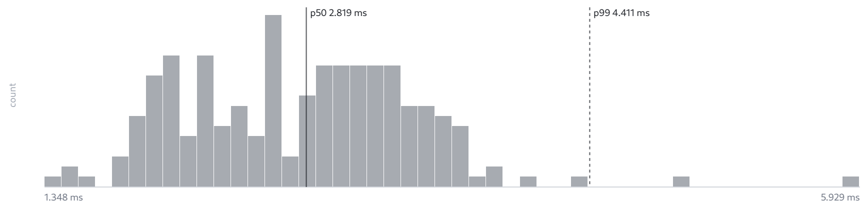
n=200 · p50 279.66 µs · p99 379.22 µs · max 397.25 µs

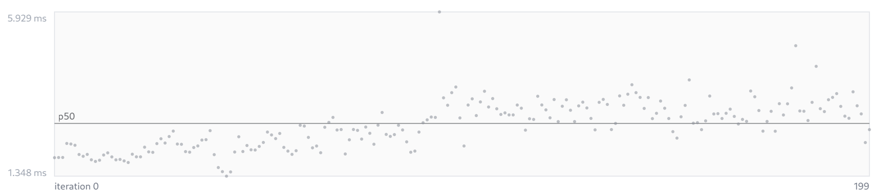




total

n=200 · p50 2.819 ms · p99 4.411 ms · max 5.929 ms





Benchmark chart

Notes and Caveats

- **Why P-256, not Ed25519.** Cosmian's PKCS#11 provider's `key_algorithm_from_attributes` function does not recognise the bare `Ed25519` `CryptographicAlgorithm` KMIP attribute (only `AES`/`RSA`/`EC`/ `ECDH` are handled), so Ed25519 private keys created via `ckms` are invisible to the provider and unusable for signing. This is tracked upstream as Cosmian/kms#1183.
- **Signature format.** `CKM_ECDSA` on this provider returns a DER-encoded, variable-length ECDSA signature (up to 72 bytes for P-256) over a SHA-256 digest of the message — not a raw fixed-length `r || s` pair, and not over the raw message. `sign-tx` presizes its buffer to the maximum DER length and hashes client-side before calling `C_Sign`, preserving the single-round-trip design.
- **Not a broadcastable Solana transaction.** Solana account addresses are 32-byte Ed25519 public keys; a P-256 public key is a 65-byte SEC1 point, so the benchmark's "from" field is truncated/ padded to fit the wire shape. The resulting transaction is not valid or broadcastable, which is irrelevant here — `sign-tx` never broadcasts, and only the signing payload shape and timing matter for this measurement.
- **Latency, not throughput under load.** Signing is sequential, one session, one thread — these percentiles describe per-signature latency, not what a concurrent/pipelined client would observe under load.

Raw Data

Raw per-iteration samples: `samples.csv` (200 rows, columns: `iteration`, `sign_init_ns`, `sign_ns`, `verify_ns`, `total_ns`).